

Stacked Adjectival Modification and the Structure of Nominal Phrases

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This article is a preliminary attempt to look at adjectival ordering restrictions (henceforth AOR) and aspects of the syntax of stacked adjectives across several languages¹ within a broad Principles and Parameters (P&P) framework. I will be examining the notion of “fixed order” and proposing that it be considered part of Universal Grammar (UG). As a working strategy, I will attempt to explore the hypothesis that adjectival distribution follows the pattern of adverbial distribution along the lines of Cinque’s *Universal Hierarchy of Clausal Functional Projections* (Cinque 1999, 106). I will argue for a framework that treats adjectives not as adjuncts, but as specifiers of distinct functional projections that are intrinsically related to aspects of their semantic interpretation. At present, attributive adjectives are usually analyzed as recursively stacked adjunct specifiers of extended A' /N' or even recursive NP projections. Yet if all adjectives had the same categorial status, the syntax, being unable to distinguish the ungrammatical from the grammatical strings, would inevitably generate fragments like **a red heavy good table*. Following Zwarts (1992), this article will attempt to demonstrate how the standard lexical/functional dichotomy may be approached through a rich semantic ontology, and how the semantic universe may be studied via an independently motivated system of syntactic categories.

1. Problematic nature of stacked adjective research

It is fair to say, both within “traditional” linguistics and generative grammar, that very few linguists have looked at the issue of AOR in any detail.² Sproat and Shih

(1988, 486) call it a “hoary problem”; Hetzron (1978, 175) considers it one of “the most delicate topics in linguistics.”

There are various reasons for this. First, there is no clear agreement in the literature as to the fundamental questions of (1) just how many semantic categories of adjective there actually are; (2) what order these categories are generated in; and (3) whether or not that order is universal. Moreover, various other syntactic/semantic and perhaps even pragmatic criteria come into play when dealing with stacked adjectival ordering. For example, there is the issue of “comma” intonation:³ asyndetically coordinated APs exhibiting intonation that “breaks the adjectives up into separate intonational phrases” (Sproat and Shih 1988, 465) thereby destroying what Hetzron call the adjectives’ “cohesion.” All writers distinguish between those strings that exhibit comma intonation (as in *a luxurious, hot, steaming bath*) and those that are spoken “without a break” (such as *a bright red balloon*). Comma intonation is an example of Sproat and Shih’s (henceforth S&S) parallel direct modification.⁴ In the preceding example, *luxurious*, *hot*, and *steaming* all modify the head noun directly, yet they also all modify the noun in parallel—together at the same time. Confusingly, however, parallel direct modification as defined by S&S need not necessarily be an example of comma intonation, and, similarly, their hierarchical direct modification⁵ need not involve the semantic categories of adjectives that are normally considered to take part in AOR. Sproat and Shih (1991, 592) give the example of *senatorial industrial investigations* and *industrial senatorial investigations* from Levi (1975). In the first example, *industrial* directly modifies *investigations* and *senatorial* directly modifies the hierarchical structure [[*industrial*] *investigations*]. As S&S note, both orderings have different meanings:

[The first] refers to the senate’s investigations of industry whereas the second refers to industry’s investigation of the senate. These semantic differences follow, as Levi shows, from the differing assignment of the deverbal nominal head’s thematic roles to the modifying adjectives. . . . Hence these ordering facts are tied to the thematic structure in a way in which the kinds of ordering restrictions which we will be investigating are not. (S&S 1988, 467 n.1)⁶

In other words, “in these cases there is no basic ordering; rather the ordering chosen depends upon the intended interpretation” (S&S 1991, 592).

Second, there is the issue of adjective combinations that would normally exhibit AOR, but where an adjective has been preposed for reasons of topic or focal stress, as in *red big ball*. Given that any XP can be preposed to TopicP/FocusP, it is hardly surprising that we find adjectives that have been similarly fronted. This article, then, will concern itself mainly with those cases of unmarked, hierarchical direct modification, as defined by S&S, without overt comma intonation as applied to result and simple, noncomplex event nominals. In other words, orderings that are unmarked and basic “in the sense that one uses the prescribed orders *unless* one intends a special interpretation” (S&S 1991, 592 n.2).

Following is a short critique of past analyses.

As I have argued previously (Scott, 1998a, chap. 1), a major problem with the majority of essays that have discussed AOR in the past is that either they take the hierarchical ordering restrictions proposed in the literature for granted or they illus-

trate only those clear cases of ordering restrictions between a few easily definable categories. Thus Hetzron assumes an extended version of Dixon's (1982) hierarchy, and while it is true that he does refine Dixon's classification somewhat, he nevertheless fundamentally accepts both Dixon's ordering restrictions and his semantic categories. S&S discuss only clear-cut cases of AOR: QUALITY > SIZE > SHAPE > COLOR > PROVENANCE. The "murkier" categories (TEMPERATURE, AGE, PATTERN) are not even mentioned. It is fair to say that none of the literature has questioned these generally accepted ordering hierarchies or even asked how many categories actually exist.

Furthermore, all writers⁷ maintain that AOR can be adequately accounted for using broad "psychological" criteria, yet none of them are able to provide a convincing argument—which is, moreover, consistent with the data—for a psychological basis to AOR. Among these accounts, it is Hetzron's whose is the least convincing. He states that "the major rule is to place the more objective and undisputable qualifications closer to the noun, and the more subjective, opinion-like ones farther away" (Hetzron 1978, 178), and a large part of his article is spent trying to give evidence in support of this claim. Yet, to give just two examples, the category COLOR comes before the category ORIGIN in linear ordering because judgments of color are supposedly open to more disagreement than are judgments of nationality. Yet can one seriously argue that this is the case? Surely, if I show you a [*red Russian ball*], the ball's color is immediately more apparent than its origin. With those categories that require more subtle gradations of judgment, Hetzron's arguments sound even slightly tenuous. To account for the order in [*a long thin blade*], Hetzron writes that "thickness requires more careful observation, and is hence more reliable as a judgement, than length, a dimension that is too easily perceivable and is therefore taken more lightly" (1978, 180). Presumably, one must conclude that in this latter example Hetzron is arguing there would be more disagreement about a blade's length than its thickness.

S&S's cognitive account suffers from the same drawbacks as those found in Hetzron; as S&S state at the end of their 1988 article (p. 486), "we have not as yet, however, described the link between the cognitive basis of the ordering restrictions and their linguistic manifestation; why should more 'apparent' adjectives be ordered closer to the head than less 'apparent' adjectives? . . . Unfortunately, we can give no answers at the present time." So the accusation still stands that it is impossible to show conclusively that it is indeed the case that the more *apparent* an adjective is, the further right in the adjective string it goes. In their 1991 article, S&S suggest that AOR can be accounted for by stating that restrictions must apply only over those hierarchical strings that directly θ -mark their sisters; but, as they are forced to admit, this is an ad hoc solution since "it is not clear from what general principles of grammar such a stipulation would follow and it is not clear what phenomena besides AOR such a stipulation would account for" (S&S 1991, 591–592).

In both articles, S&S similarly do not question the standard assumed ordering hierarchies. Indeed, like Androutsopoulou (1996), they take as given the fact that AOR occur, and they are mainly interested in proposing possible structural configurations for the various cases where stacked adjectives do or do not display AOR. Theoretically, S&S propose that stacked adjectival modification "involve[s] iterative adjunction of adjectives to N" (S&S 1988, 473). So, in English, where adjectives modify an NP, the structure would be:

- (1) [_{NP} rapid [_{NP} Russian [_{NP} invasion of Afghanistan]]] (S&S 1991, 571)

If adjectives are adjoined iteratively to the head noun, how can S&S account for the fact that they display ordering restrictions? A syntactic analysis as in (1) will of course generate a *rapid Russian invasion*, but it will also generate fragments like **a new red bright dress*. Although they are adequate as far as dealing with the differences between those cases where multiple stacked adjectives do and do not display AOR, the configurational structures presented in both S&S and Androutsopoulou are not able, as far as the cases that *do* display AOR are concerned, to account for and predictively generate the actual ordering restrictions themselves.

2. Stacked adjectives and X' -theory

Stacked adjectives, like stacked adverbs, are generally analyzed in X' -theory as recursive adjunct projections. Such an analysis is not without its problems, however. To begin with, there is evidence that adjectives (like adverbs) are not adjuncts, since one of the basic criteria for judging adjuncts (Freidin 1992, 43) is that their ordering appears to be free:

- (2) a. a book [about physics] [in German]
b. a book [in German] [about physics] (Freidin 1992, 43)

Similarly, relative clauses also display free ordering:

- (3) the man [who paid the bill] [wearing a stripy scarf] [of medium build] [with a gold tooth] (Cook and Newson 1996, 145)

Example (3), containing two relative clauses and two PP adjuncts, can be ordered in any of its possible combinations without focusing any of its elements. This cannot be done with either stacked adjectives or stacked adverbs, since even if a nonstandard order is still grammatical, one of the elements will receive a definite focal/contrastive stress:

- (4) a. John repeatedly viciously attacked Bill.
b. *John viciously repeatedly attacked Bill.

(5) a. Mine's the big red car.
b. *Mine's the red big car.

Here, the (b) examples are "ungrammatical" unless *viciously/red* is said with focus (i.e., emphatic or contrastively stressed) intonation.

An analysis in which adjectives are adjoined iteratively to the head noun is inadequate because it is unable to account for the fact that adjectives display clear ordering restrictions. Moreover, this type of structural analysis is theoretically undesirable as it would make NPs unique in X' -theory: no other category allows recursive adjunction to an intermediate projection. Abney recognizes the same fact when he writes that such an analysis not only "espouses adjunction in the base, [but] furthermore, adjunction to a non-maximal category. An embarrassing question is why there are no elements adjoined to any other single-bar projection at d-structure: not V', P', A', I', C', etc." (Abney 1987, 323).

A problem with recent analyses such as Cinque (1994) and Giusti (1997) is that, even though in these works stacked adjectives are analyzed as specifiers of functional heads, they are not *distinct* functional heads. Apart from the lexical/functional difference, theirs are essentially the same type of adjunction analysis as the standard X' treatment; consequently, of course, they will not be able to take into account the fact that adjectives occur in AOR. Moreover, Giusti's proposed structural configuration (see Giusti 1997, 97) seems to suggest that whole AgrPs may be recursively projected. Under this analysis, adjectives (and presumably adverbs) would again be unique in X'-theory as they would be the only functional phrases capable of being iteratively stacked—DPs, for example, cannot further project into DPs. Indeed, such an analysis would seem to run counter to Grimshaw's (1991) thesis of "extended projections."

2.1. A short summary of Cinque (1999)

Within the background of the current proliferation of research into functional projections (FPs), Cinque (1999) proposes a correlation between adverbials and semantically related FPs that are associated with tense, aspect, modal force, and so on. As Cinque himself states, the aim of his work is to justify the hypothesis that adverb phrases (AdvPs) are the unique, overt manifestations of the specifiers of distinct maximal projections (and *not* recursive IP/VP adjuncts) and to argue for the existence of a fixed universal hierarchy of clausal FPs. He does this by producing evidence for a functional-semantic hierarchy of AdvPs across a range of languages; by independently producing evidence for a hierarchy of clausal functional-semantic projections, again across a range of languages; and then, finally, by amalgamating both hierarchies and showing that there is a systematic one-to-one mapping between the AdvPs and the clausal functional heads to which they are associated. Cinque concludes: "The transparent semantic relation holding between each adverb class and the contiguous head morpheme to its right (when the two hierarchies are matched) has been taken to suggest that each AdvP is the specifier of the phrase projected by the corresponding functional head morpheme" (Cinque 1999, 140).

Cinque's hypothesis, then, is that each adverbial class must be occurring as the specifier of a phonologically null or "empty-headed" FP to which it is semantically related, the order of adjectives at S-structure/Spell-Out being a direct mapping from the ordering of the FPs themselves. As an example, and to closely paraphrase Zanuttini (1997, 228)—who discusses an earlier version of Cinque's work in some detail with respect to negative markers and movement of nonfinite verbs in Romance—Cinque is able to show how adverbs belonging to the class "already" occur as specifiers of a temporally related projection that expresses the relationship between the event time and reference time; how adverbs of the class "no more" occur in an aspectual projection expressing perfectivity; and how past participles occur cross-linguistically in any head position made available by an FP that has an adverb as its specifier (i.e., in all the head positions labeled "e" in the following examples). For example:

- (6) [_{TP2}already [_T e [_{AspP}_{perf} no more [_{Asp} e [_{AspP}_{gen/prog} always [_{Asp} e]]]]] (based on Zanuttini 1997, 228)

Cinque suggests that there may be somewhere in the region of forty FPs that are present in all languages and proposes that, even where a particular projection is not overtly realized morphologically, it is still present structurally in the syntax.

Finally, it should be noted that while correlations between adverbs and FPs associated with, for example, aspect and tense have been proposed before in the literature, Cinque's work (and his *Universal Hierarchy of Clausal Functional Projections* in particular) stands out as the most comprehensive and clearly articulated cross-linguistic statement on the topic to date.

2.2. Extending Cinque's thesis to adjectival projections

If, as is generally accepted, "the structure of DP is essentially parallel to that of CP," and that "NP is the nominal counterpart of VP, and DP is the nominal counterpart of CP" (Valois 1991, 12 and 34, respectively), it is a natural and logical step to go from looking at FPs associated with the clause to looking at FPs associated with the DP. Although there is little functional morphology in the DP, because adverbs are the sentential counterpart of adjectives, we might expect to find similar FPs hosting adjectives. Indeed, this notion has already been proposed in the literature⁸ but (1) either the nature of the FPs has been left unspecified (as in Cinque 1994, who proposes a general FP, "F") or (2) the FPs are limited to certain types of nominal (i.e., event nominals).

As noted earlier, positing an analysis in which stacked adjectives are the specifiers of FPs of particular semantic classes has one distinct advantage over the traditional adjunction analysis: it predicts ordering restrictions. However, there are other conceptual and empirical reasons for adopting the stance advocated here. A number of these are also given in Cinque (1994):

1. The linear ordering of each stacked adjective can be viewed as a direct mapping from the hierarchical ordering of the FP with which it is associated.
2. Why does there seem to be a limit to the number of adjectives found within the DP (according to Cinque 1994, 96, "apparently not exceeding six or seven")? A priori there need be no such limit and, indeed, the adjunction hypothesis would predict that no such limit exists. The generation-in-Spec analysis at least in part predicts why: the number of adjectives is limited by the semantic possibilities for modification of the number of projections found between D and NP.
3. A theory allowing only Spec-Head relationships could presumably be said to be more "minimalist" in spirit than one that countenances both specifiers and adjuncts.⁹
4. Within the traditional adjunction hypothesis, the fact that APs appear to the left of their head has to be specifically stipulated in the phrase structure rules (and, of course, the existence of multiple Specs is allowed), whereas Kayne's (1994) antisymmetric view of syntax predicts there is always only one single specifier per projection and that that specifier must be left-branching¹⁰—thus, under the present assumptions, Kayne's abstract theory receives further empirical support.

5. An analysis in which adjectives are the specifiers of FPs associated with their respective semantic classes leads to a tighter, more articulated correspondence between universal semantic properties and the syntax than in the recursive adjunction approach.
6. Cinque's (1994) N-raising account suggests independent evidence that FPs exist within the DP, between DP and NP.
7. In many languages, stacked adjectives display case and ϕ -features, so within a minimalist view of syntax (Chomsky 1995a), adjectives can only be in Spec positions: the Spec-Head relationship is the only structural configuration where they can get their features "checked" and erased¹¹ since only specifiers are within the checking domain of a head; adjuncts are not.
8. (Tentatively) FPs are projected from closed lexical classes (e.g., determiners, pronouns, and classifiers); if adjectives are semantically related to the functional heads of the various FPs in the Spec of which they find themselves generated, then we might expect some "adjectival" interpretation or function to sometimes be realized as a closed class item. In some languages, at least, this does indeed seem to be the case.
9. Finally, if AOR are a direct and overt manifestation of the ordering of FPs, then conjectures as to the psycholinguistic motivation for AOR need not be posed: AOR fall out as a direct consequence of UG.

Most of the preceding arguments, speak for themselves, I think. However, I would like to elaborate briefly on two points. First, as mentioned tentatively in point 8, "adjectives" do indeed seem to form a closed class in some languages. Dixon states that "not all languages have the major word class Adjective. Either they have no Adjective class at all, or else there is a small non-productive minor class that can be called Adjective"; he also notes that "languages that have only a limited class of adjectives show considerable similarity in the concepts that are expressed through adjectives" (Dixon, 1982, 2 and 3, respectively). So, for instance, he discusses the cases of Igbo with only eight adjectives; Hausa with about twelve; and the Australian aboriginal language Malak Malak with seven.¹² Interestingly, the adjectives that these languages possess fall into (one or two members of) the same semantic categories—usually SIZE, LENGTH, AGE and COLOR. Dixon shows (1982, 7) that out of twenty languages he surveyed which had "closed" adjective classes, *large* occurred in all twenty languages; *small* in nineteen; *short* in fifteen; and *long*, *old*, *bad*, and *white* in fourteen. The second, minor, issue I wish to note is that, with regard to point 9, an obvious advantage of positing that AOR are part of UG is that the complexity of the learning procedure necessary for a child to master the restrictions is minimized (and the "logical problem of language acquisition" answered), and the linguist need not have recourse to such difficult-to-define psychological notions as S&S's *amount of comparison/computation* (S&S 1988, 467–472) or *absoluteness* (S&S 1991, 588–591).

Finally, due to the poverty of nominal functional morphology, the existence of functional heads related to the semantic classes associated with AOR is practically impossible to motivate independently within the DP (in the way that Cinque 1999 is able to look at clausal FPs related to AdvP classes *independently* of the classes themselves). It may be, however, that the augmentative/diminutive affixes found in the

derivational morphology of adjectives in languages like Russian and Portuguese are overt manifestations of functional heads within the “AP.” In Russian we find such pairs as *byeliy* ‘white’ with its diminutive form *byel-yen’k-iy* ‘nice and white’ (the incorporated diminutives are in boldface italics in these examples) used to express endearment; and *bol’shey* ‘big’ with its augmentative counterpart *bol’sh-ooshch-iy* ‘very large’ used to express awkwardness or ugliness.¹³

One could argue that these are overt manifestations of a SIZE-related functional head; similarly, classifier systems in those languages that have them may be the overt manifestations of a SHAPE-related FP. In a similar vein, Cinque (1994, 106 n.27) speculates whether adjectives like the Italian *attuale* ‘present’, *passato* ‘past’, and *futuro* ‘future’ are instantiations of FPs within the DP related to TenseP and AspectP. Hockett (1958, 238) describes the existence within the nominal system of Potawatomi, an Algonquian language of North America, of a morphologically overt manifestation of the category Tense or Aspect that is identical to that found within the Potawatomi verbal system: viz. *nkāšatās* ‘I am happy’ (a verb) and *nčīman* ‘my canoe’ (a noun) versus *nkāšatsapən* ‘I was formerly happy (but am not now)’ and *nčīmanpən* ‘my former canoe (now lost, destroyed or stolen)’. And, of course, it is quite possible that there exist other languages in which such DP-internal functional heads are not empty.

3. Adjectives and semantic classes

Since adjectives are the specifiers of FPs, what actual FPs are they the specifiers of? If adverbs are, as Cinque suggests, the specifiers of such FPs as Mood_{speech act}P, Asp_{proximate}P, T(Future)P—that is, of the FPs that reflect the semantic classes by which they pattern in linear ordering—it seems reasonable, and by Uniformity indeed theoretically desirable, to assume that adjectives, their nominal counterparts, are likewise treated as the specifiers of FPs that reflect the semantic classes according to which they are ordered.

This entails looking at the classes of adjectives themselves, to research just which adjectives belong to which class; exactly how many classes of adjectives there are; whether certain adjectives can belong to more than one class (as is the case with adverbs); and finding out whether these classes and the ordering restrictions in which they occur are found cross-linguistically. A full discussion of all these questions is naturally out of the scope of an article this size: all I can hope to do is highlight some of the questions that need answering in future research.

Because strings with more than three stacked adjectives at once (for example Hetzron’s 1978, 179 *a happy young blind Belgian sheep dog*) are comparatively rare and moreover quite long, they tend to make it difficult to judge which of the various possible orderings is the most common or most natural. Fragments of this sort allow greater freedom mathematically for elements to be preposed or focused: thus, *a young happy blind Belgian sheep dog*; *a blind happy young Belgian sheep dog*; *?a young blind happy Belgian sheep dog*, and so on, are all possible. With this in mind, I shall therefore build a proposed hierarchy using two, maximally three, adjectives at a time. The AOR list I shall be using as a basis will be that of Kingsbury and Wellman:¹⁴

- (7) DETERMINER > SUBJECTIVE COMMENT > SIZE > AGE > SHAPE > COLOR > NATIONALITY/ORIGIN
> MATERIAL > COMPOUND ELEMENT > NOUN (1986, 40)

Kingsbury and Wellman's (henceforth K&W) cline of categories is able to deal with orderings such as the following (where any other ordering in these languages is decidedly marked):

- (8) a. *suuri vanha musta englantilainen koira* (Finnish)
 big old black English dog
 b. *idiak akaan ndaidat ofong* (Ibibio)
 ugly old red dress
 c. *etak ekara okpokoro* **ekara etak okpokoro* (Ibibio)
 small round table *round small table*

Their cline of categories suggests a relatively rich functional hierarchy. Yet there is evidence that (1) some of these categories could be further decomposed and (2) there may well be other categories not even mentioned by them. For example K&W's category cline is unable to account for the following ordering restriction (which appears to be SIZE > TEMPERATURE):

- (9) *pitkä kuuma kesä* **kuuma pitkä kesä*¹⁵ (Finnish)
 long hot summer *hot long summer*

As I have previously suggested, it is necessarily out of the scope of an article this size to give full and adequate attention to all the questions posed here. Instead, I shall limit myself to illustrating how complex the issue is by attempting to formulate an answer to the following three questions:

1. Can the category *SizeP* be further decomposed?
2. Is there a *SpeedP*?
3. Is there a *TemperatureP*?

Each of these three questions, if discussed extensively, could be the topic of a full-length article on its own. Moreover, there are, of course, theoretically other possible categories (e.g., DEPTH, STRENGTH) that I am not discussing. Thus I repeat, I simply wish to highlight some of the complexities involved.

3.1. Refining the standard stacked adjective ordering hierarchy

As Hinton and Marsden (henceforth H&M) note (1985, 80), it seems that the category SIZE can indeed be "split" into at least two more categories HEIGHT and WIDTH, with HEIGHT preceding WIDTH:

- (10) a. *pitkä lihava mies* **lihava pitkä mies* (Finnish)
 tall fat man *fat tall man*
 b. a short skinny woman *a skinny short woman
 c. a low narrow ceiling *a narrow low ceiling
 d. *uyarnna kattiyulla bhithi* **kattiyulla uyarnna bhithi* (Malayalam)
 high thick wall **thick high wall*

Within the umbrella category of SIZE, in addition to adjectives related to width and height, are also adjectives related to length and weight. These, too, seem to follow AOR. To begin with, it seems that LENGTH precedes all other categories:

- (11) a. a long high cliff face *a high long cliff face
 b. dugačka uska ulica *uska dugačka ulica (Serbo-Croat)
 long narrow street *narrow long street*
 c. lyhyt ohut terä *ohut lyhyt terä (Finnish)
 short thin blade *thin short blade*
 d. a long heavy canoe *a heavy long canoe
 e. pitkä korkea laiva *korkea pitkä laiva (Finnish)
 long tall ship *tall long ship*
 f. haf hir poeth *haf poeth hir¹⁶ (Welsh)
 summer long hot *summer hot long*
 g. en lång vid kjöl *en vid lång kjöl (Swedish)
 a long wide skirt *a wide long skirt*

Similarly, it appears that WEIGHT follows all other categories; in particular, it seems to follow the category WIDTH:¹⁷

- (12) a. visoka teška statua *teška visoka statua (Serbo-Croat)
 tall heavy statue *heavy tall statue*
 b. paksu painava kirja *painava paksu kirja (Finnish)
 fat heavy book *heavy fat book*
 c. kanatha kanamkuranja bukə *kanamkuranja kanatha bukə (Malayalam)
 thick heavy book **heavy thick book*
 d. a slim light volume *a light slim volume

So it would seem that the FP SizeP can be further split into the following functional phrases: LengthP > HeightP > WidthP > WeightP.¹⁸ This hierarchy would consequently predict that a string such as *a long tall narrow ship* with any other ordering would produce a slight focal stress (and indeed this seems to be the case).

Dixon (1982, 24) suggests that SPEED should be considered a separate adjectival semantic class and that it should be placed after the category he calls PHYSICAL PROPERTY (which includes *hard, heavy, hot, rough*) but before the categories HUMAN PROPENSITY (*wicked, clever, happy*) and AGE. Certainly it seems that adjectives of the category SPEED (*fast, slow, quick*) do take part in AOR:

- (13) a. ɔsɔp ufa moto *ufa ɔsɔp moto (Ibibio)
 fast new car *new fast car*
 b. ein schnelles altes Pferd *ein altes schnelles Pferd (German)
 a fast old horse *an old fast horse*
 c. afon araf lydan *afon lydan araf (Welsh)
 river slow wide *river wide slow*
 d. ein langsamer breiter Fluss *ein breiter langsamer Fluss (German)
 a slow wide river *wide slow river*

But where in the ordering hierarchy should this category come? The following data from English may suggest a positioning:

- (14) a. a long fast road *a fast long road
 b. a long slow river *a slow long river¹⁹

- c. a tall fast ship *a fast tall ship
 d. a low fast plane *a fast low plane

At first sight, it may seem that SPEED comes after the class SIZE. Note, however, the following:

- (15) a. dugačka spora reka *spora dugačka reka (Serbo-Croat)
 long slow river *slow long river*
 b. hidas levä joki *levä hidas joki (Finnish)
 slow wide river *wide slow river*
 c. spori teški tenk *teški spori tenk (Serbo-Croat)
 slow heavy tank *heavy slow tank*
 d. tanc araf trwm *tanc trwm araf (Welsh)
 tank slow heavy *tank heavy slow*

Interestingly, it seems that the fact adjectives belonging to the class SPEED engage in AOR lends direct support to the hypothesis that the semantic class SIZE should be split. It appears, at least in the above languages and in English, that adjectives from the class SPEED come after adjectives of LENGTH and HEIGHT, but before adjectives of WIDTH and WEIGHT. If the semantic class SIZE were a single, homogenous class, then these facts would be difficult to explain. This then suggests a further refinement to the category cline: LengthP > HeightP > SpeedP > WidthP > WeightP.²⁰

Finally, as the data in (9) indicate, there seem to be definite ordering restrictions with regard to adjectives from the class TEMPERATURE, at least as far as English and Finnish are concerned. This pattern seems to be repeated across other languages (see (11f) for Welsh):

- (16) a. ein langer heißer Sommer *ein heißer langer Sommer (German)
 a long hot summer *a hot long summer*
 b. dugo toplo leto *toplo dugo leto (Serbo-Croat)
 long hot summer *hot long summer*
 c. niinda thanutha varshakālam *thanutha niinda varshakālam (Malayalam)
 long cold rainy season *cold long rainy season*
 d. en lång kall vinter *en kall lång vinter (Swedish)
 a long cold winter *a cold long winter*

Thus it seems as if one might wish to include a *TemperatureP* within the hierarchy of FPs. Yet where exactly would a *TemperatureP* come? Evidence from English (and the above languages) suggests after SUBJECTIVE COMMENT and adjectives related to SIZE, but before SHAPE:

- (17) a. a beautiful long hot summer *a beautiful hot long summer
 b. ?a heavy hot round pan *a hot heavy round pan
 c. a cold square lump of ice *a square cold lump of ice

However, within K&W's (and all other) category cline, the category AGE also comes between SIZE and SHAPE. This means that either the categories AGE and TEMPERATURE are interchangeable (as is suggested by Fowler 1992, 138) or these categories display AOR with respect to each other. In fact, the latter conclusion seems to be the case (although, probably for semantic reasons, it is uncommon for adjectives from the classes TEMPERATURE and AGE to come together):

- (18) a. an extremely hot old oven *an extremely old hot oven
 b. a red-hot Victorian marble fireplace *a Victorian red-hot marble fireplace
 c. a cold young puppy *a young cold puppy

If AGE and TEMPERATURE had displayed interchangeable orderings that did not affect focus, then we could reasonably have expected them to be members of the same category (or of a category AGE /TEMPERATURE).²¹ Since this does not seem to be the case, I suggest that they do belong to separate FPs with the order AGE > TEMPERATURE. Thus our final, enriched (but still incomplete)²² hierarchy of projections looks as follows:

- (19) *A proposed universal hierarchy of AP-related functional projections (a first approximation)*
- | | | | | | | | |
|---------------|-------------------------|----------------------|--------------------------|---------|--------------------------------|----------|--------------------|
| [DP | <i>the/this</i> | [Subj.CommentP | <i>hasty/magnificent</i> | [SizeP | <i>big/small</i> ²³ | [LengthP | <i>long/short</i> |
| [HeightP | <i>tall/high</i> | [SpeedP | <i>fast/slow</i> | [WidthP | <i>fat/thick/thin</i> | [WeightP | <i>light/heavy</i> |
| [TemperatureP | <i>hot/cold</i> | [AgeP | <i>Edwardian/modern</i> | [ShapeP | <i>round/triangular</i> | | |
| [ColourP | <i>dark brown/white</i> | [Nationality/OriginP | <i>German/Parisian</i> | | | | |
| [MaterialP | <i>rosewood/brass</i> | [NP | ... | | | | |

3.2. Some theoretical consequences of this approach

If we take a minimalist framework like Chomsky's *Bare Phrase Structure* model (Chomsky 1995b), then we can assume that phrasal categories are not distinguished by their level and that heads and maximal projections are all the same category; the only difference being that a nonbranching X is a head and that "a category that does not project any further is a maximal projection[:] . . . any other is an X'" (Chomsky 1995b, 396). Thus an adjective is an AP in the Spec of a semantically related FP; it is both an X^o (a head) and an X^{max} (a phrase). This has the intuitive advantage that adjectives can still be treated as fully lexical elements and so therefore have lexical meaning (consequently possessing all the ambiguity and exceptionality²⁴ associated with the lexicon—see sections 4.1 and 5.1 for respective discussion on the adjectives *cool* and *good* in particular); at the same time, this model allows for the FPs, in whose Spec each adjective is generated, to determine both the adjectives' hierarchical syntactic ordering and their combinatorial semantic interpretation. Such an analysis accords well with one of P&P theory's fundamental tenets that "lexical heads provide the descriptive content; . . . functional heads determine the configurational geometry . . . thus contributing to the determination of the interpretation as well as the form of the linguistic expression" (Belletti and Rizzi 1996, 3). It also accords well with standard versions of the *Minimalist Program*, whereby all FPs are headed by a grammatical (in this case predominantly semantic) feature bundle and all elements are "generated" in lexical projections (LPs) and "licensed" in FPs (see Zwart 1997, 250): stacked adjectives are generated as lexical maximal projections (APs) and are licensed by hierarchically ordered FPs that are intrinsically related to aspects of their semantic interpretation.²⁵

An important consequence of this approach is that adjectives can have a "core" meaning, but that same core meaning may be interpreted differently according to which FP the adjective has been generated under: thus one could argue an adjective like *old* has been generated alternately in an AGE-related FP in a fragment like *the old*

man with the walking stick, in some kind of TEMPORAL-related FP in *my old* (=former) *boss* or *an old* (=ancient) *joke*, and even in an adjunction structure within a SUBJ. COMMENT-related FP in a phrase such as *good old Fred* (whereby it is adjoined to the adjective *good*). Likewise, an adjective may belong to a “core” semantic class, but which class it is *interpreted* as belonging to depends on the semantic interpretation of the head it is associated with (thus, the same adjective at different times seems to appear to belong to different classes).

It has to be noted that an adjective is not the overt, lexical instantiation of an FP per se (this is also the case, respectively, with adverbs and clausal FPs for Cinque). FPs are not adjectival (or even lexical) in nature themselves: they are instantiations of bundles of “functional” (=syntactico-semantic) features. What is important is that an FP is a projection that can host in its Spec any element related to its semantic interpretation or field: APs, AdvPs, NPs, PPs, and, of course, morphological affixes (related to tense, aspectual features, etc.). A consequence of this is that it enables us to provide structures for the following:

- (20) a. an off-the-cuff remark
b. a ready-to-eat meal

To take [*an off-the-cuff*] *remark*] as an example, there is clear evidence through coordination and modification phenomena that *off-the-cuff* behaves syntactically and semantically as an adjective:

- (21) a. We found his remarks rather [_{AP} off-the-cuff] and [_{AP} uninteresting].
b. His remarks were both [_{AP} off-the-cuff] and [_{AP} weird].
c. His remarks were ridiculously/extremely/very/too [_{AP} off-the-cuff].

Yet it also seems to behave like a PP, as only PPs can be modified by *straight* and *right* (i.e., *straight/right down the road*):

- (22) a. His remarks were straight [_{PP} off-the-cuff] and [_{PP} to-the-point].
b. His remarks were right [_{PP} off-the-cuff].
c. *His remarks were straight [_{AP} uninteresting].
d. *His remarks were right [_{AP} uninteresting].

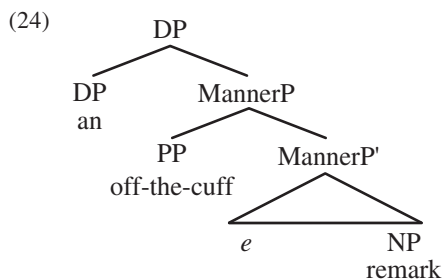
But in X' -theory Ds do not take PPs as complements so, in the case of [*an off-the-cuff*] *remark*] and morphologically complex “adjectives” like it, for theoretical reasons we probably do not want to claim that this is a special or unique instance of a D being followed by a PP. If we assume for now that *off-the-cuff* somehow has dual AP/PP status, then structurally this AP/PP must nevertheless be in the Spec of some phrasal category—be it AP or PP:

- (23)
-
- ```

graph TD
 DP1[DP] --- DP2[DP
an]
 DP1 --- XP[? XP]
 XP --- PP[PP
off-the-cuff]
 XP --- Xp[? X']
 Xp --- X[? X]
 Xp --- NP[NP
remark]

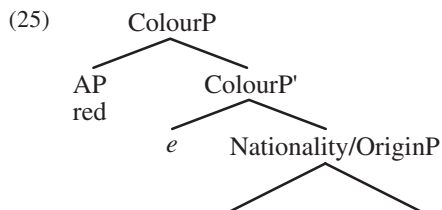
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Under Abney's (1987) analysis of nominal modification, *off-the-cuff*, being headed by the preposition *off* has to be analyzed as a PP that somehow "inherits" AP status presumably from its NP complement. With the analysis presented in this article, together with *Bare Phrase Structure* assumptions, *off-the-cuff* remains a PP but receives its semantic interpretation from the FP it finds itself in the Spec of (presumably some kind of MannerP):



In much the same way, we can also preserve the distinction between nominal modification and the adjectival category in such phrases as *summer desserts*, *law school regulations*, *music teacher*, and *night bus* in which the nominal can remain an NP and is simply an NP in the Spec of an adjectival FP from which it receives its interpretation.<sup>26</sup>

Finally, this approach has one further advantage over Abney (1987) who analyzes [*red cars*] as an AP. If we assume, as in current theory, that two elements from the Numeration are concatenated by a transformation Merge that projects the features of only one out of the two elements (Chomsky 1995b, 397), intuitively, I think, we would want the head/lexical features of the noun to be projected since this fragment denotes a set of entities. If the lexical head features of the adjective were projected as in Abney's approach, this projection would incorrectly predict that the resulting larger structure be interpreted as a property—that is to say, a predicate. The structure in (25), however, merely states that the simple noncomplex noun in *red cars* has an extended projection into a non-event functional hierarchy. This analysis allows the adjective to be a head *and* an AP, but since it is the specifier of an FP, which is simply an extended projection of grammatical features from the noun, the noun itself can still be viewed as the head of the phrase because (unlike in Abney) it has not projected into a projection that is lexical:



#### 4. Semantic classes and DP-internal functional projections

The comment made in passing during the previous section about ordering interchangeability and semantic class deserves some elaboration. As far as I am aware, this was first noticed by Dixon who writes that “there appears to be an underlying order between types but not within types” and who concludes that the interchangeability of *[a brave clever man]* with *[a clever brave man]* is proof that they belong to the same category (1982, 25).<sup>27</sup> This fact was used here in conjunction with the AOR displayed by TEMPERATURE and AGE adjectives to show that they belong to different classes. Within a class, it does indeed seem that AOR do not occur:

- (26)    a. egy okos bátor ember      egy bátor okos ember    (Hungarian)  
               *a clever brave man*          *a brave clever man* (Hetzron 1978, 171) (both SUBJECTIVE COMMENT)
- b. a thick fat book              a fat thick book (both WIDTH)
- c. a new modern kitchen        a modern new kitchen (both AGE)

If it *is* the case that adjectives from the same class are interchangeable,<sup>28</sup> this fact and the fact that adjectives are specifiers of the FPs of the semantic classes in which they group together make certain predictions:

1. Adjectives with the same orthography but which can occur in different positions must be able to be specifiers of more than one (i.e., different) FP.
2. Conversely, adjectives (like *ancient* and *old*) that appear to belong to the same semantic category but whose ordering facts suggest otherwise must therefore be specifiers of different FPs.

#### 4.1. Same adjective, different FPs

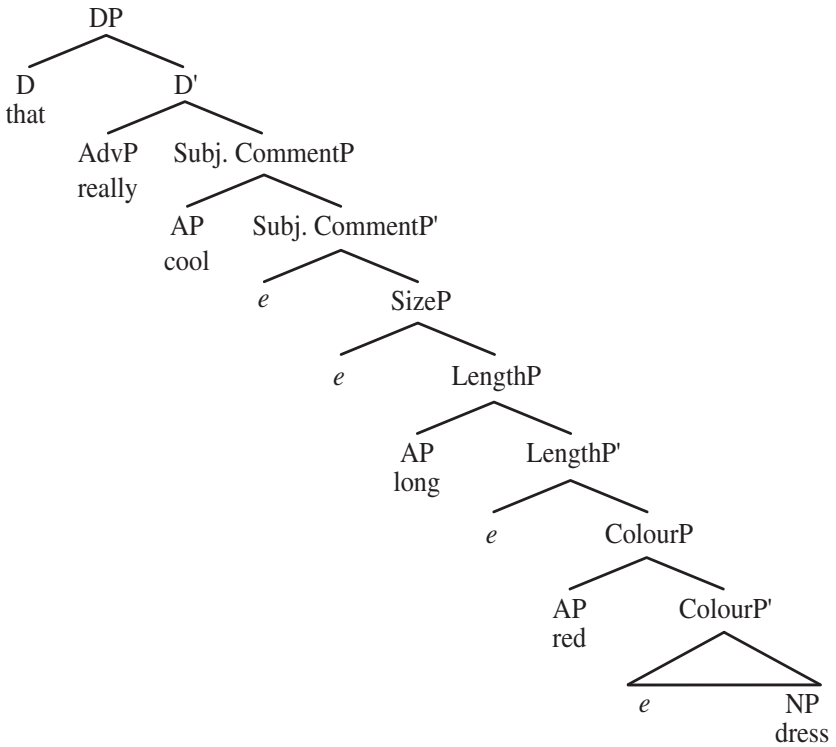
The first point highlights yet again the semantico-syntactic similarity between adjectives and adverbs, once more suggesting the same syntactic analysis for both classes. Take the following (now classic) facts from Jackendoff (1972, section 3.2):

- (27) a. John has answered their questions cleverly.  
b. John cleverly has answered their questions.  
c. John has cleverly answered their questions.  
d. John has cleverly answered their questions cleverly/stupidly.

Cinque (1999, 19–22) shows how the adverb *cleverly* is base-generated in three different structural positions in the tree and receives different semantic interpretations, depending on where it has been generated.<sup>29</sup> Cinque argues that since “a restricted theory should force a one-to-one relation between position and interpretation (i.e., one specific, and distinct interpretation for each position of ‘base generation’),” (1999, 20), it is empirically and theoretically preferable to locate each of the three *cleverlys* as the specifiers of distinct, semantically related FPs (so, for example, the *cleverly* in 27b would be the specifier of a Mod<sub>ability/permission</sub>P). Applying this

idea to adjectives would account for the fact that an adjective's interpretation is a function of where in the tree it has been base-generated. For example, I would argue a phrase such as [*that really cool long red dress!*] has the following structure:

(28)



Intuitively, *cool* is not the specifier of a TemperatureP here: it does not have the semantic reading *not hot*. Instead, as the specifier of a Subj.CommentP, it receives the interpretation *great* or *excellent*. Some syntactic evidence shows this to be the case:<sup>30</sup>

- (29) a. What a long cool red dress!  
 ["ungrammatical" where *cool* receives the interpretation *excellent*]  
 ["grammatical" where *cool* receives the interpretation *not hot*]  
 b. What a cool long red drink!  
 ["ungrammatical" where *cool* receives the interpretation *not hot*]  
 ["grammatical" where *cool* receives the interpretation *excellent*]

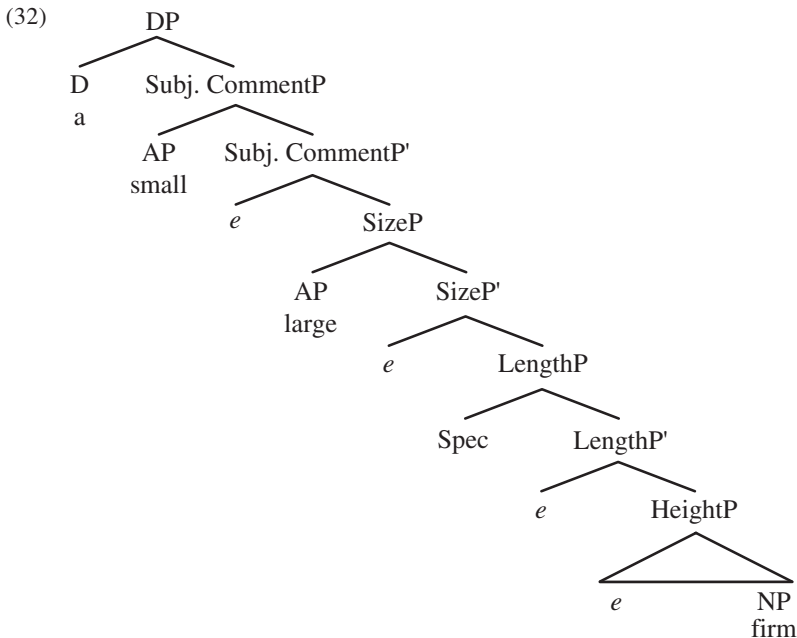
If we follow the hierarchy of FPs suggested in (19), it becomes clear why *cool* in (29a) cannot receive the interpretation of *great* or *excellent*—it has, in fact, been generated in the Spec of a TemperatureP and therefore receives the interpretation "cool to wear." Similarly, *cool* in (29b), having been generated in a Subj.CommentP position, is unable to mean "not hot" and is forced to take on a "subjective comment" reading. Thus this analysis provides a syntactic account for an area of the grammar that is usually viewed as the domain of the lexicon (the dual readings for *cool*) or even pragmatics, and it also illustrates clearly how the semantic universe may be studied via theoretical syntax.

I would argue that the preceding analysis is also able to account for the following data (often discussed in the literature and seen as a “problem” for the syntax and so left to the semantic component of the grammar to deal with):

- (30) a. a small large firm  
b. a large small firm

- (31) he is a green young writer

It is interesting to note that there is no way that (30a) and (30b) can be instances of S&S’s parallel modification, since this would lead to a contradiction (*??a firm which is small and large*), so it must be that they are both instances of direct modification. Furthermore, the fact that in the fragment [*a small large firm*], *small* directly modifies [*large [firm]*] is shown by the semantics: the firm is large, but small for a large firm. How can this reading be accounted for syntactically? The analysis here suggests an answer. In (30a), for example, *large* is generated closer to the head noun within SizeP and takes on a size-related interpretation, and *small* is generated further up the tree within Subj.CommentP and takes on a subject comment interpretation (and vice versa, of course, for [*a large small firm*]):



Likewise with [*a green young writer*], according to the hierarchy in (19), *green*, when indicating COLOR is generated after AgeP; here, it has been base-generated *before* AgeP in Subj.CommentP and has the meaning *inexperienced*. Thus we are able to account for the difference in meaning between *he is a young green* (=color) *Martian* and *he is a green* (=inexperienced) *young Martian* and, similarly, between *an ugly cold* (=temperature) *woman* and *a cold* (=heartless) *ugly woman*. Furthermore, this suggests why the standard word order is *he is a young black writer*: *he is a black*

*young writer* seems to give the impression that *black* means “depressive,” “macabre,” or even “bad” (*black* has been base-generated away from its usual position in the structure, higher up in Subj.CommentP). Thus I would suggest adjectives like *cool* and *good* are only genuinely “ambiguous” in the lexicon; the semantic reading they receive within the sentence does not depend on the grammar’s pragmatic component but is, rather, a direct function of where the adjective has been syntactically generated within the AOR structural hierarchy I have proposed.

#### 4.2. Same apparent semantic class, different FPs

The idea that adjectives are the specifiers of distinct FPs can also account for what would otherwise be the odd patterning behavior of adjectives such as *ancient* and *old*, which, although apparently belonging to the same semantic class AGE, have very different syntactic distributions:

- (33) a. He arrived in an ancient old car.  
 b. \*He arrived in an old ancient car.  
 c. a tiny little man  
 d. \*a little tiny man

Why is the ordering in (33b) ungrammatical? If we assume that *ancient* can only be a specifier of Subj.CommentP, then, consequently, (33b) should only be grammatical with a stressed or contrastive reading (i.e., with “old” stressed). Why should *old* and *ancient* belong to different categories? As a tentative answer, it may well be that certain adjectives (*young/old* for AgeP; *big/small* for SizeP; *heavy/light* for WeightP) form the semantic “core” meanings of their respective FPs<sup>31</sup> and that others, although related semantically, may either be found only in Subj.CommentP (as in *ancient*; *bulky*, *ponderous*; and probably *tiny*) or have “dual” status in that they may occur in their “core” FP and also in Subj.CommentP (although without taking a stress or contrastive reading).<sup>32</sup> Thus, we would predict *ancient* and *old* are not interchangeable. And this prediction seems to be borne out:

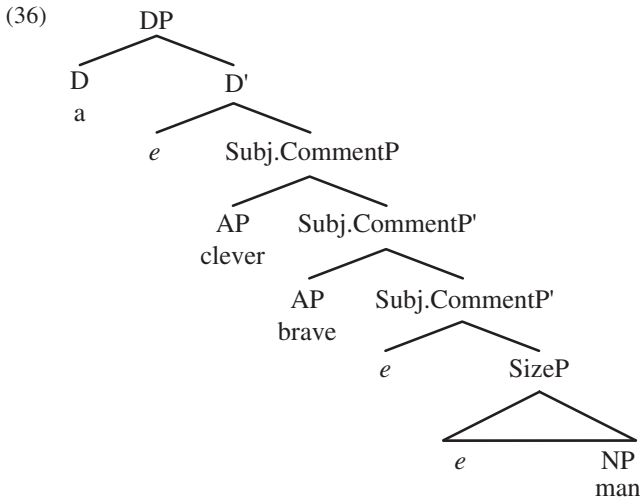
- (34) a. a long wide old river  
 b. ??a long wide ancient river [but, an ancient long wide river]

#### 4.3. Same semantic class, interchangeable ordering

The question now arises as to how one should analyze stacked adjectival phrases that really *are* composed of adjectives from the same category. For example:

- (35) a. a boring famous book vs. a famous boring book (both Subj.CommentP)  
 b. a new young recruit vs. a young new recruit (both AgeP)

One could propose that [*a clever brave man*] for example, consists of recursively stacked Subj.CommentPs. However, as we have seen, this analysis is not to be preferred since stacked adjectival phrases would then be the only complete XPs in X'-syntax that are capable of being recursively stacked. One way, I suggest, is to consider cases of this sort as genuine adjuncts, recursively adjoined to intermediate projections of the head (as in the traditional analysis of stacked adjectives):



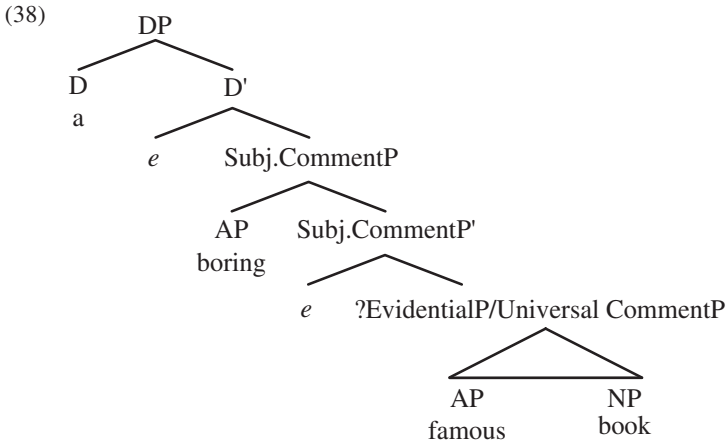
Since they are projections of the *same phrasal category*, the adjectives here may be stacked hierarchically and recursively, with order being unimportant (*a clever brave / brave clever man*); and if, as is usual, we take it that the computational system proceeds in a “bottom-up” fashion, the adjunct (here, *clever*) will modify the structure to which it is adjoined, [*brave [man]*]. This would also give theoretical credence to Dixon’s (1982) hitherto unmotivated observation that adjectives from the same category seem to appear in any order.

One disadvantage of this analysis is that throughout this article I have argued extensively against an adjunction approach—and yet in this subsection I appear to be seeking to advocate exactly what I have been trying to disprove! Therefore an alternative way of analyzing the data in (35), and one that is in keeping with the generation-in-Spec analysis proposed throughout this work, is to see if the various “same categories” can themselves be further decomposed. Take, for instance [*a boring famous book*] and [*a famous boring book*], which, under the preceding analysis, are both simply instances of adjoined Subj.CommentPs. Such an analysis does not predict that the fragments can have the following semantic interpretations:

- (37) a. Oh, it’s a really boring famous book.  
 [“grammatical” with the interpretation “a book generally/universally considered famous but that I find boring”]  
 [“ungrammatical” with the interpretation “a book I find famous but that is generally/universally considered boring”]  
 b. Oh, it’s a really famous boring book.  
 [“grammatical” with the interpretation “a book generally/universally considered boring but that I consider nevertheless famous as (or, for being) a boring book”]  
 [“ungrammatical” with the interpretation “a book generally/universally considered famous but that I consider nevertheless boring as (or, for being) a famous book”]

It seems the category SUBJECTIVE COMMENT can be further decomposed into (a) some sort of Evidential or Universal Comment Phrase that indicates some kind of

general, stable, or standard evaluation and (b) a phrase that can be considered more essentially “subjective comment” in that it indicates the speaker’s own evaluation or an evaluation that may be more transitory:



An interesting topic for further research would be to see if the preceding analysis could be extended to cover strings like *a clever brave / brave clever man* and other such “same category” combinations—for example, *a thick fat / fat thick book* (both WIDTH) and *a modern new kitchen / a new modern kitchen* (both AGE).<sup>33</sup>

## 5. General and specific topics for future research

Like the final paragraph of the previous section, this section suggests some areas of possible future research vis-à-vis AOR.

It seems that, unlike Cinque’s *Universal Hierarchy of Clausal Functional Projections*, the overall hierarchy of FPs found within the DP, while almost certainly being universal (at least with respect to hierarchical direct modification), may not be universal as far as the specific fine-grained ordering of the FPs is concerned. For instance, it could be that some languages (e.g., Finnish) may allow parametrization in the ordering hierarchy.<sup>34</sup>

- (39)
- |    |                         |                                                                   |
|----|-------------------------|-------------------------------------------------------------------|
| a. | an ugly long road       | pitkä ruma tie<br><i>long ugly road</i>                           |
| b. | beautiful long red hair | pitkät kauniit punaiset hiukset<br><i>long beautiful red hair</i> |
| c. | beautiful big red ball  | suuri kaunis punainen pallo<br><i>big beautiful red ball</i>      |

According to my informants, the orderings in (39) are the most natural and unmarked, whereas the orderings corresponding to the English seemed to them to bear focal stress. Thus in Finnish it may be either that LengthP/SizeP precedes Subj.CommentP in the FP hierarchy, or the adjective *pitkä* ‘long’, for example, is a Subj.Comment adjective in that language.<sup>35</sup> The fact that there may be parametric

variation within limited boundaries should come as no surprise since parametric variation within the IP has been noted by a number of linguists (for example Ouhalla 1991) and the very locus of cross-linguistic parameterization is now thought to be exclusively the domain of FPs. If parameterization exists in the hierarchies of DP, AgrP, and NegP, why not between the FPs found broadly within the “AP”?

Similarly, Hetzron reports the phenomenon of what he terms, “big”-fronting in Amharic, Japanese, Spanish, and Romanian: the placing of the adjective *big* (but not usually *little*) before all other adjectives, as in the following example:

- (40) *təlləq qonǵo qäyy kʷas* (Amharic)  
*big pretty red ball*  
 ‘pretty big red ball’ (Hetzron 1978, 172)

If this particular example is indeed direct hierarchical modification,<sup>36</sup> then some languages may place SizeP before Subj.CommentP. Indeed, according to my own data (Scott 1998a, 72), it seems that Serbo-Croat may indeed be such a language. In either case, it appears that, just as in variation within the IP, these parameterizations are quite specific: only one or two categories vary (AgrP and TP within the IP, for example; and LengthP and Subj.CommentP within the DP).

Another area for future research that has been mentioned at various points throughout this article is the number and nature of FPs found within the “AP.” Other areas include questions such as the following: Do groups of adjectives that belong to the same semantic class display similar overall syntactic behavior?<sup>37</sup> Do we really need (for example) a category PHYSICAL DEFECT (*blind, deaf*), as Hetzron proposes? Is *long in a long new book* a LengthP or a Subj.CommentP? Should we only assign the central semantic terms to the semantic classes being proposed (*heavy/light* for WeightP; *long/short* for LengthP), so that the less “central” the term is, the more it is open to being seen as SUBJ. COMMENT (*hefty, bulky, cumbersome, ponderous*). And, of course, there are many classes of adjective that I have not had space to deal with: Dixon’s HUMAN PROPENSITY class (*jealous, wicked*), for example,<sup>38</sup> and participles used as adjectives (*freckled, lumbering, computerized, aged*), to name but a few.

### 5.1. FPs and complex internal event structure

Another issue not dealt with in this article is the question of adjectives displaying AOR that do not belong clearly to any of the semantic categories in (19). Valois (1991, chap. 4) for example, discusses how adjectives that have adverbial equivalents display distribution patterns cross-linguistically within event nominal projections that are identical with their adverbial counterparts:

- (41) a. They [<sub>TP</sub> probably [<sub>VP</sub> completely invaded Jupiter] ].  
 b. \*They completely probably invaded Jupiter. (Valois 1991, 149)
- (42) a. the probable complete invasion of Jupiter  
 b. \*the complete probable invasion of Jupiter (Valois 1991, 151)

According to Valois, these data are identical in French. He is led to conclude that “adjectives in event nominals, like adverbs in clauses, belong to different classes, and as such are generated at various places in the DP,” and he suggests the hierarchy

is “*probable-class* > *frequent-class* > *complete-class*” (Valois 1991, 175, 165). As to what these classes are, what other members they have, and where they fit in the hierarchy I have proposed is, I can only say again, outside the scope of this article, but it is tempting to speculate that there may turn out to be a close correlation between Cinque’s *Clausal Functional Projections* and the FPs under which adjectives that display AOR group.

In this chapter I have examined only AOR for simple and result nominals like *dog* and *exam* and not for complex event nominals like *criticism*, *examination* and *criticizing*; *examining*—that is to say, nominalizations in general.<sup>39</sup> A question I posed earlier was why there should be fewer FPs in the DP than in the CP. I suggested this may be related to the simple/event nominal dichotomy: *a beautiful big red ball* is OK, but not *\*a beautiful big red invasion*; similarly, to use Valois’s example, *the probable complete invasion* is again perfectly fine, but not *\*the probable complete ball*. Since simple *R* argument nominals are not associated with an internal aspectual event structure, it is not surprising that much of the articulated aspectual and temporal FPs that characterize the CP are missing from projections that extend from nouns with *R* arguments. Does this mean there are two sets of functional hierarchies: one for simple/result nominals; the other for event nominals that incorporate an aspectual event projection? Certainly, according to Grimshaw, the external argument for complex event nominals is different from the external argument for simple and result nominals: “The possibility that the external argument of a complex event nominal is not *R* but something else offers potential insight into the differences between complex event nominals and the others. Since determiner and modifier systems are different for the two kinds of nouns, this can be attributed to their different external arguments” (Grimshaw 1990, 66). She calls the external argument associated with complex event nominals “*Ev*.” Does this difference in external argument between the two types of nominal force a different set of extended functional AP-related projections and therefore a different set of AOR: one for *R*- and one for *E*-type nominals?<sup>40</sup> Grimshaw states, “no noun with *R* as its external argument can ever have event structure associated with it. . . . It is event structure that licenses modifiers like *frequent* and *constant*” (1990, 67). Can such projections be mixed (*?the big frequent investigating*; *?the ugly constant criticizing*) or integrated into a single, all-encompassing functional hierarchy? Again, I can only point to these questions as worthwhile areas for future research.<sup>41</sup>

Two other sets of data also deserve to be mentioned. To begin with, in [*the good good typist*],<sup>42</sup> where the sense is of a good (=fast, efficient) typist who is also good morally (as opposed to [*the morally bad good typist*]), the *good* nearest the head noun is constrained to receive the reading *good in ability* (=fast), whereas the *good* that modifies the structure [*good [typist]*] can only receive the reading *morally good*. Thus the following reading is ungrammatical: *\*the good* (in ability) *good* (in morality) *typist*; similarly, *a bad good typist* cannot mean “a typist who is bad at typing but, morally, a good person.” This data, I would argue, is further proof that an adjective generated in the hierarchy relates directly to the semantic reading it receives. Why else should it be that *good* in the sense of “morally good” can only be attributed to the first adjective in the linear series? Naturally this poses a further problem for the analysis presented here in that, in *a good good typist*, where do the two *goods*

come in the hierarchical sequence? The “outer” *good* seems clearly to belong to Subj.CommentP, but what about the “inner” one nearest the head noun?<sup>43</sup> It may be that it is generated in some kind of MannerP.

## 5.2. Adjectives and focus structures

Throughout this article I have argued that it cannot be the case that adjectives are adjuncts that can simply be recursively stacked. Andrews uses the following data to argue (the traditional argument) that “modifiers are introduced by nesting rules expanding N' to N' plus a preceding or following modifier” (1983, 696).

- (43) a. an alleged English baron  
b. an English alleged baron

What Andrews fails to notice, it seems, is that (a) and (b) differ quite clearly with respect to markedness: (b) is by far the more marked version, with *English* receiving contrastive and focal stress. A syntax that assigns both fragments the same structure is missing out on an important difference between the two.<sup>44</sup> Within the analysis suggested in this article, (43a) would be assigned a structure as follows:

- (44) a. [<sub>DP</sub> an [<sub>SpecSubj.CommentP</sub> alleged [ [<sub>SpecNationality/OriginP</sub> English [<sub>NP</sub> baron ] ] ] ] ]

And (43b) would consequently have a derivation something like:

- (44) b. [<sub>DP</sub> an [<sub>FocusP</sub> English<sub>j</sub> [<sub>SpecSubj.CommentP</sub> alleged [ [<sub>SpecNationality/OriginP</sub> t<sub>j</sub> [<sub>NP</sub> baron ] ] ] ] ] ]

Since I have argued throughout this article that recursively stacked adjectives are not adjuncts but XPs, then, like any lexical XP, they should be capable of being preposed into some sort of FocusP.<sup>45</sup> This analysis has the advantage of (1) treating adjectives like any other XPs (i.e., capable of being preposed) and (2) reflecting in a more articulated way the tight correspondence that exists between the syntax and semantics when it comes to AOR. Thus all of the following could be said to be instances of focus preposing:

- (45) a. It's the green<sub>i</sub> big t<sub>i</sub> chair that I want.<sup>46</sup>  
b. It's the old<sub>j</sub> green<sub>i</sub> big t<sub>j</sub> t<sub>i</sub> chair that I want.  
c. It's the nice green<sub>i</sub> big t<sub>i</sub> chair that I want.  
d. It's the nice green<sub>i</sub> old<sub>j</sub> big t<sub>j</sub> t<sub>i</sub> chair that I want.

Note also:

- (46) Carol's horrible six children made life miserable for her second husband. (Andrews 1983, 697)

To account for this last example, Andrews argues “it seems that QP is also introduced by a recursive rule expanding N' to QP N'.” Again, what Andrews seems to miss is that *Carol's horrible six children* is a FocusP (hence the scope differences which he himself comments that “it is the group of six children rather than their cardinality that is horrible” [1983, 697]). Thus if Carol had twelve children, it would be quite possible to say, *Carol's horrible six children made life miserable for her second husband, but her nice six gave him great joy*. To posit this as an example of preposing to FocusP is, I would argue, a much simpler and more elegant solution,

which, moreover, has the advantage of providing an analysis that unifies this phenomenon with other Focus phenomena not involving QPs.

## 6. Conclusion

In this article I attempted to look at the issue of AOR and the semantic classes into which adjectives displaying AOR group. Although the approach has been within a broad P&P framework, to integrate the linguistic data with the analysis outlined it has been necessary to also focus on aspects of the semantics of AOR. Consequently, I highlighted the extent to which the semantic universe may be studied via an independently motivated system of syntactic categories, and, in doing so, I stressed the fact that there exists a tighter interaction between the syntactic and semantic components of the grammar than is usually thought. In this article I argued that previous analyses of AOR were in general not sufficiently articulated and proposed the following refinement to the current ordering hierarchies found in the literature:

- (47) *A proposed universal hierarchy of AP-related functional projections for noncomplex and result nominals (a second approximation)*  
 DETERMINER > ORDINAL NUMBER > CARDINAL NUMBER > SUBJECTIVE COMMENT >  
 ?EVIDENTIAL > SIZE > LENGTH > HEIGHT > SPEED > ?DEPTH > WIDTH > WEIGHT >  
 TEMPERATURE > ?WETNESS > AGE > SHAPE > COLOR > NATIONALITY/ORIGIN >  
 MATERIAL > COMPOUND ELEMENT > NP

Furthermore, on the basis of cross-linguistic data, I suggested that both non-complex and complex event nominal projections have a hierarchically articulated functional structure similar to that which Cinque (1999) proposed for the clause. The thesis presented—whereby adjectives are the specifiers of functional projections with which they are semantically associated—has been shown to explain many aspects of the cross-linguistic syntactic ordering restrictions, as well as semantic combinatorial interpretation of adjectival modifiers. Thus, in this chapter I explain how such an analysis can be extended to accommodate cases of modification by other XP categories (for example, NPs and PPs).

## Notes

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1. See Scott (1998a) for more cross-linguistic examples.
2. The main essays dealing with AOR are Dixon (1982), Hetzron (1978), the work of Sproat and Shih (1988 and 1991), and Androutsopoulou (1996).
3. Also referred to as “comma disjuncture” (Dixon 1982, 24).
4. See Sproat and Shih (1991, 568) for discussion.

5. See Sproat and Shih (1991, 567–568).

6. This is related to Grimshaw's (1990) simple versus event distinction for nominals, to which I return briefly in section 5.1. In her terminology, nouns divide into two classes: complex event nominals that have both an event and an argument structure, and others (noncomplex, simple event nominals and result nominals) that have neither. On an intuitive level, "non-argument-taking nouns refer to individuals and simple events and argument-taking nouns refer to complex events" (Grimshaw 1990, 57). In this article I primarily examine the modification of simple and result nominals: "simple" nominals (like *computer* and *dog*) and result nominals (like *exam*; see Grimshaw 1990, 49) that have an external nonthematic argument *R*, but no external thematic argument *E* (see Higginbotham 1985; Grimshaw, 1990, 63)—that is, nouns that are not linked to aspectual semantics. Levi's (1975) example *investigation* is clearly a complex event nominal.

7. Excluding Androutsopoulou (1996) who—rightly, I think—argues that AOR must be part of UG.

8. See Cinque (1994) and Crisma (1990). Indeed, according to Cinque (1994, 97), Crisma has proposed at least seven Spec positions for APs: [<sub>DP</sub> D [<sub>XP</sub> possessive X [<sub>YP</sub> cardinal Y [<sub>WP</sub> ordinal W [<sub>ZP</sub> speaker-oriented Z [<sub>HP</sub> subject-oriented H [<sub>NP</sub> [thematic/manner] N]]]]]]].

9. According to Zwarts (1992, 18 n.5), apparently Hoekstra (1991) argues that all specifiers *are* adjuncts, but adjuncts that agree with their head. Adjectives obviously seem to fit with this analysis. Indeed, it could be argued that, under this definition, adjectives would be prototypical specifiers.

10. Indeed, Kayne states that "adnominal adjective phrases that are not complements must then be specifiers, each of a separate head. . . . The same must hold for adverb phrases" (1994, 137 n. 31; see also *idem*, 100–101).

11. Or where the relevant features of the head noun can "percolate" up, depending on how one views the checking strategy: see Radford (1997, 158) for a percolation account.

12. See also Tallerman (1998, 34 and 44) for elementary discussion of the fact that adjectives form a closed class in many languages. In addition, there are languages where the semantic equivalents of "adjectives" are members of the verbal class, and it is debatable whether such languages have a syntactic category Adjective (i.e., lexical items with the feature matrix specification [+N,+V]) at all. In Japanese, for example, "adjectives," being in reality stative verbs, not only display tense and aspectual morphology but also do not display AOR (this is the indirect modification patterning discussed by S&S; modification in such languages is typically the same as for relative clauses and appositives). Lakhota (Van Valin and LaPolla 1997, 28) is apparently another such language.

13. Data from Pulkina (n.d., 350–351).

14. I must stress that there is no particular reason why I have chosen to use this list other than it is relatively extensive to begin with and the category definitions are quite clear as to what they delineate. In general, the hierarchy lists found in the English-language teaching (ELT) literature are more fully articulated than those in the linguistics literature. For a full summary of the various hierarchy lists proposed throughout the linguistics and ELT literatures, see Scott (1998a, appendix 1).

15. The grammaticality judgments of many of the following examples are, in places, extremely subtle. The asterisk, \*, does not necessarily indicate "ungrammaticality" as such but reflects the fact that some of the readings are simply more marked or focused.

16. Adjectives in VSO languages like Irish and Welsh follow English ordering (and are *not*, as one would expect, the mirror-image of English) because there is movement of the NP past the adjectives: presumably the NP crosses over the adjectives in a cyclic movement in order to assign case to the DP. See S&S (1988, 485) for a brief explanation. Head fronting is frequently used to explain various syntactic phenomena in the Celtic languages (for ex-

ample, sentence-initial placement of the verb and the noun phrase placement of the nominal head).

17. Intuitions involving the category WEIGHT are, to me at least, probably the most difficult and subtle to judge.

18. As far as I am aware, this is the first time this ordering has been proposed.

19. Examples like *a slow long river* particularly highlight the difference between comma intonation (in which this fragment is grammatical: *a slow* [pause], *long* [pause], *river*) and non-comma intonation.

20. One of the categories also occurring within SizeP (which I will not be discussing) is probably a DepthP: *a long deep trench* (?*a deep long trench*); *a deep wide ravine* (\**a wide deep ravine*)—note that these facts seem to hold for Swedish as well. The question when proposing new categories, of course, is where does one stop? Thus the adjective *wet* displays definite cross-linguistic ordering restrictions (*a cold wet day* versus \**a wet cold day*); does this really mean one should go so far as to posit a “WetnessP” for example(!!!)?

21. See section 4 for discussion of adjective classes and ordering interchangeability.

22. Again I repeat that this is not the full hierarchy. A complete hierarchy would include a Cardinal/OrdinalP, a NumberP, and so on to account for orderings such as *the last few years*, *the first three men*, and *the latest Paris fashions*. See, for example, Ritter (1991, 50) whose work discusses the functional category NUMBER with respect to Hebrew.

23. Although (due to lack of space) they are not discussed in this section, it is clear that adjectives belonging to the core category SIZE (*big*, *small*) precede all other size-related categories: *big fast*, \**fast big*; *big long*, \**long big*; *big tall*, \**tall big*; *big wide*, \**wide big*; *big hot*, \**hot big*.

24. Chomsky (1995a, 235) refers to the lexicon as merely “a list of ‘exceptions.’”

25. Within a minimalist framework, it may be that the very fact that stacked adjectives are the specifiers of semantically related FPs falls out as part of the general Principle of Full Interpretation (Chomsky 1986, 98). Since each element of the sentence must play some role—whether syntactic, semantic, or phonological—stacked adjectives might be the lexical instantiations of functional, DP-related semantic information that has been projected into the syntax according to the subcategorization properties of the head noun. Incidentally, it should be noted that throughout this article I will assume (as Cinque does for his *Hierarchy of Clausal Functional Projections*) that the full number of FPs is always structurally present (just sometimes empty) in every sentence.

26. Jackendoff (1997, 230 n.9) criticizes Lieber (1992), who apparently classifies similar constructions as adjectives (Jackendoff gives the examples *a Charles-and-Di syndrome*; *a let's-get-on-with-it attitude*). He writes, “she [i.e., Lieber] presents these to be adjectival compounds. But she presents no evidence that they are A°s, and I suspect they are not, since they resist modification typical of adjectives such as ??*I've never seen a more let's-get-on-with-it attitude than yours*, \**John's attitude is very let's-get-on-with-it*” (Jackendoff, 230 n.9). To me at least, both sentences are perfectly acceptable. Moreover, the analysis presented in this section would be able to deal with instances like [*a more let's-get-on-with-it attitude*]: they would presumably be full IPs generated in the Spec of (like *off-the-cuff*) a MannerP.

27. Of course, it may be that they are instances of S&S's parallel modification. I leave this issue open. Hetzron (1978, 169; 171) illustrates the fragment *a brave clever / clever brave man* across a number of languages and concludes that Dixon's statement seems to hold.

28. And it is by no means conclusive that they are: compare *a small little trinket*, \**a little small trinket*; *an old Victorian kitchen*, \**a Victorian old kitchen*.

29. This is a “pure manner” interpretation (John answered in a clever manner) in 27a; a “subject-oriented” interpretation (it was clever of John to answer their questions) in 27b; and an ambiguous reading (ambiguous between the two possible readings) in 27c.

30. These are the interpretations when both sentences are said with normal unmarked stress. It might be possible (although I think unlikely) for *cool* to receive the interpretation *excellent*, of course, if *long* in 29a is said with focal stress.

31. So *heavy* is much more of a prototypical WeightP adjective than, say, *bulky*. The notion of a certain X having more “X-ness” or being a much more representative “X” than a comparable and similar X is found right across cognitive science as a whole. To quote Fodor: “even concepts like ODD NUMBER, which clearly do have definitions, often have prototype structure as well. The number 3 is a ‘better’ odd number than 27. . . . The discovery of the massive presence of prototypicality effects in all sorts of mental processes is one of the success stories of cognitive science” (1998, 93).

32. It is hard to think of adjectives that might fit this description, however.

33. A related topic is how, and in what larger semantic groups, such FPs themselves cluster: so, for example, Cinque’s full *Universal Hierarchy of Clausal Functional Projections* (Cinque 1999, 106) shows how FPs within the clause group themselves (broadly!) into Mood-, Tense-, Modality-, and Aspect-related projections. A similar phenomenon might be found with AP-related FPs. For instance, the evidential-, evaluative-, and “subjective comment”-related FPs discussed in this section group themselves in the outermost projections furthest from the head (just as in Cinque’s analysis for the clause, Mood speech act-, evidential-, and evaluative-related FPs occur outermost). Within the DP, next come all the projections related to size and physical shape, perhaps paralleling clausal projections related to Aspect—almost as if within the DP, physical shape and size are manifested in projections hosting adjectives whose semantics reflect the “concrete” nature of the nominal (*round, long*) and, within the clause, temporal “shape” and “size” are manifested in Aspectual projections hosting adverbs whose semantics reflect the internal abstract “shape” of events (*often, briefly*).

34. An interesting question for further research is whether there are any other languages like Finnish. Hetzron (1978, 173) notes a similar phenomenon: the interchangeability of SIZE and SUBJ. COMMENT (in his terminology, EVALUATION) in Chinese (although the example he gives is of indirect, so-called, DE modification—see the work of S&S—so, following S&S, interchangeability is actually to be expected), Polish, and Chichewa. In Malayalam, according to Asher and Kumari (1997, 129), it seems that adjectives from the class AGE tend to precede those of the class SIZE.

35. Another interesting question is whether there are any other adjectives like *pitkä* (perhaps *short*?) in Finnish—if all “core” adjectives of the LENGTH class precede SUBJ. COMMENT, then we have good grounds for positing parameterization; if it appears that just *pitkä* does, then we are probably dealing with a lexical peculiarity.

36. As far as the Japanese and Spanish examples are concerned, it is not. Hetzron presents the following data for a *pretty big red ball*, neither of which, in the terminology of S&S, is hierarchical direct modification:

- (i) *ookii kirei na akai booru* (Japanese)  
*big pretty red ball*
- (ii) *un balón grande rojo y lindo* (Spanish)  
*a ball big red and pretty* (Hetzron 1978, 172)

The Japanese *ookii* ‘big’ and *akai* ‘red’ are, in fact, stative verbs acting as relative clauses (i.e., “a ball *which is big, which is red*”) which correspond only semantically to their English adjectival equivalents; likewise, the *na* found after the (true) adjective *kirei* ‘pretty’ is a form of the copula obligatorily needed after adjectives in Japanese to modify the head noun, and as such it is directly equivalent to the Chinese DE modification discussed by S&S (1988 and 1991). As for the Spanish data, this seems to me to be an example of comma intonation fol-

lowed by what I have termed “*and-use*” (see Scott 1998a, 7 n.8), a usage found across all Romance languages.

37. Pustejovsky (1995, 23) discusses Dixon’s AOR list but notes, for example, that, within Dixon’s class DIFFICULTY, both *easy* and *difficult* behave syntactically as tough-movement predicates, whereas within Dixon’s class QUALIFICATION, *likely* allows raising but *probable* and *possible* do not. He concludes that Dixon’s classification system based on the notion “semantic field” is unable to predict the syntactic behavior of its members.

38. Dixon (1982, 17) suggests they come after SPEED but before AGE.

39. After Chomsky (1970), the first type of nominal (*criticism*; *examination*) is termed derived nominalization; and the second (*criticizing*; *examining*), gerundive nominalization. Grimshaw (1990) shows how both types of nominal have an inherent “event” structure, which is an internal aspectual analysis: gerundive nominals always have a-structure, and derived nominals are ambiguous. Such nominalizations are further “verb-like” in that they take obligatory arguments; simple and result nominals do not (Grimshaw 1990, 47). Certain modifiers (for example, *frequent*) force an event interpretation onto the nouns they modify; they also modify the actual internal “event” referred to by the nominal.

40. In turn, this notion would have consequences for other parts of the grammar—for example, the Projection Principle: “Representations at each syntactic level are projected from the lexicon, in that they observe the subcategorization properties of lexical items” (Chomsky 1981, 29). In theory, then, *R* argument nominals would subcategorize for a different set of FPs to *Ev* argument nominals.

41. Another area of adjectival modification that I have not had space to deal with is the case of (1) languages where certain adjectives change their semantics, depending on whether they precede or follow the noun, and (2) the syntax/semantics of such adjectives when they are stacked; cf. the following French and Welsh data, respectively: *un roi ancien* ‘an ancient king’, *un ancien roi* ‘a former king’ (Napoli 1993, 182); and *unig blentyn* ‘an only child’, *plentyn unig* ‘a lonely child’. Presumably, an account of this phenomenon might involve combining some version of Cinque’s (1994) N-movement hypothesis with the FP analysis presented here and (perhaps) explaining movement in terms of (?semantic) feature checking.

42. These data are based (loosely) on Jackendoff (1997, 62).

43. The same problem is to be found with [*an old old Etonian*]*—*this fragment can only receive the interpretation *an old* (= in age) *old* (= former) *Etonian* and never \**an old* (= former) *old* (= in age) *Etonian*.

44. Also, in (43a) *English* is under the scope of *alleged*: as is well known [*an alleged English baron*] need be neither English nor an actual baron. In (43b), in contrast, *alleged* is within the scope of *English*: the person must be English but need not necessarily be a baron.

45. Although I am suggesting here that contrastively focused adjectives are preposed to “something like” a FocusP, the actual nature of the projection to which they are moved (whether it be FocusP, TopicP, ForceP, or other P) I will leave open. See Rizzi (1997) for discussion.

46. Liberman and Sproat use the sentence *I want the red large ball, not the blue one* to argue that AOR “are only preferences, operating in the absence of pragmatic reasons to choose a different order. It is perfectly possible to choose an alternative order . . . so it seems best to argue with the traditional view that these ordering preferences reflect some sort of natural continuum of psychological affinity, even though it seems difficult to define such a scale in a non-circular way” (1992, 165). Needless to say I disagree with this: arguments against the *natural continuum of psychological affinity* view are presented more fully in Scott (1998a); and just what could the “pragmatic reasons” be that AOR are “operating in the absence of”?

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